

# Systrophē II: Quantum Field Theory on a Tipler-Pair Background

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## Abstract

We extend the open-source SYSTROPHE package ([github.com/Zynerji/systrophe](https://github.com/Zynerji/systrophe)) from a classical-GR Tipler-pair simulator (v0.1–v0.6) to a quantum-field-theoretic laboratory (v0.7–v0.13). The new infrastructure supports: (i) renormalised stress-energy tensor  $\langle T_{\mu\nu} \rangle$  on the Lewis–Papapetrou exterior via point-splitting and the Hadamard parametrix, with exact recovery of the conformal anomaly  $K_{\text{Kretsch}}/(2880\pi^2)$  in vacuum; (ii) closed-form Atiyah–Patodi–Singer  $\eta$ -invariant on the  $\mathbb{Z}_3$  Möbius cover with verified  $\eta_0 + \eta_1 + \eta_2 = 0$  anomaly closure and Callan–Harvey inflow balance; (iii) joint Floquet quasi-energies on the  $(t, b)$  Brillouin product torus with  $\mathbb{Z}_3$  cyclic-permutation symmetry; (iv) Unruh acoustic-metric mapping ( $c^2 - v^2 = F$ ) identifying the chronology horizon as an acoustic horizon, with the gravitational and acoustic Hawking temperatures coinciding to machine precision; (v) Brown–Maclay  $\langle T_{\mu\nu} \rangle$  at a Casimir cavity in flat space with LP curvature correction; (vi) Newton–Kantorovich back-reaction solver demonstrating the linear-walk failure of naïve Picard iteration. The package now contains 406 passing tests and ships the math via a clean module-per-construction layout. We document the open speculative items as ansatz-level interpretations needing external input (cavity geometry, vacuum-selection criterion, etc.), and propose specific experimental tests in BEC sonic-horizon apparatuses.

## 1 Introduction

SYSTROPHE v0.1 implemented the closed-form van Stockum / Tipler exterior with log-periodic structure and a co-axial pair construction allowing tunable CTC interference. The first whitepaper [1] established the classical-GR mathematics: exact  $\alpha = \sqrt{4a^2 - 1}$ ,  $F(r) \propto \cos(\alpha \ln r/R + \delta)$  sinusoid, phasor-sum closure, and the structural  $\delta = \pi$  extinction. The present paper extends to QFT on this background.

### 1.1 Scope of Systrophē II

The v0.7–v0.13 series of releases adds the following independent machinery, each with a clean module and a separate test suite:

- **v0.7:** Radial Dirac operator on the LP exterior with bound-state spectrum (`dirac.py`, `dirac_spectrum.py`).
- **v0.8:** Quantum-field-theory in curved spacetime infrastructure: Dirac sea pressure, particle creation at the chronology horizon, QFTCS back-reaction trace, vacuum-energy proxy (`dirac_sea.py`, `particle_creation.py`, `qftcs_backreaction.py`, `quantum_diagnostics.py`).
- **v0.9:** Point-splitting renormalisation; full Riemann tensor, Kretschmann scalar, exact 4D conformal-anomaly trace, and DeWitt heat-kernel  $a_2$  coefficient (`point_splitting.py`).
- **v0.11:** Research-grade adiabatic Floquet on the radial Dirac problem (`floquet.py`).

- **v0.12:** Topological-Casimir mode sums on the  $\mathbb{Z}_3$  cover with the Hurwitz  $\zeta$  at  $s = -3$  closed form (`casimir.py`).
- **v0.13:** *This paper.* Eight new modules covering the post-anomaly machinery (Sections 2–5 below): `hadamard_offtrace`, `anomaly_inflow`, `floquet_mobius`, `acoustic_metric`, `casimir_throat`, `newton_kantorovich`, `tipler_fractal`, `horned_torus`; plus four extension modules `back_reaction`, `floquet_engineering`, `dsi_observables`, `adm_export`, `d_ctc`.
- **v0.15:** Six validation modules promoting previously ansatz-level claims to concrete calculations (Section 8 below): `wormhole_throat`, `exotic_matter_accounting`, `chronology_protection`, `wilson_loop`, `dynamical_casimir`, `vacuum_states`.

## 2 Renormalised stress tensor and off-trace components

The package’s exact 4D conformal anomaly is

$$\langle T^\mu_\mu \rangle_{\text{ren}} = \frac{K_{\text{Kretsch}}}{2880 \pi^2} \quad (\text{vacuum, } R_{\mu\nu} = 0). \quad (1)$$

Module `hadamard_offtrace.py` extends this to the full *tensor*

$$\langle T_{\mu\nu} \rangle_{\text{ren}} = \frac{1}{2880 \pi^2} R_{\mu\rho\sigma\tau} R_\nu^{\rho\sigma\tau}, \quad (2)$$

whose trace recovers (1) exactly. The state-independent geometric piece (2) is testable against the trace anomaly numerically; state-dependent corrections (Boulware vs. Hartle–Hawking) are documented but not implemented (they require mode-sum machinery beyond this paper).

We verify the trace-matching condition numerically:

```
trace_decomp = trace_decomposition(vs, r=2.0)
trace_decomp["trace_anomaly_check"] # < 1e-13
```

## 3 Anomaly inflow on the $\mathbb{Z}_3$ cover

The angular Dirac operator on  $S^1$  with twist  $\alpha$  has APS  $\eta$ -invariant

$$\eta(\alpha) = 1 - 2\alpha, \quad \alpha \in (0, 1), \quad (3)$$

with  $\eta(0) = 0$  (symmetric regularisation).

For the  $\mathbb{Z}_3$  Möbius cover ( $\alpha_b = b/3 + \gamma_{\text{eff}}/(2\pi)$ ), the three branches have

$$(\eta_0, \eta_1, \eta_2)|_{\gamma=0} = (0, \frac{1}{3}, -\frac{1}{3}), \quad (4)$$

and

$$\sum_{b=0}^2 \eta_b = 0, \quad (5)$$

the anomaly closure condition. A non-zero gauge twist  $\gamma_{\text{eff}}$  breaks (5); the residual is cancelled by a Chern–Simons bulk inflow with coefficient  $k_{\text{CS}} = 1/(24\pi^2)$ :

$$k_{\text{CS}} \cdot q_{\text{flux}} = -\frac{1}{2} \sum_b \eta_b(\gamma_{\text{eff}}). \quad (6)$$

This is the Callan–Harvey relation evaluated explicitly on the  $\mathbb{Z}_3$  cover; verified by direct construction in `z3_anomaly_inflow_balance`.

## 4 Acoustic-metric mapping and analog Hawking

Unruh’s 1981 acoustic-metric mapping identifies the LP angular metric as an acoustic line element via

$$\rho_{\text{acoustic}} = \sqrt{L}, \quad v_\phi = K/\sqrt{L}, \quad c^2 - v_\phi^2 = F. \quad (7)$$

The chronology horizon  $F = 0$  coincides exactly with an *acoustic* horizon ( $c = v$ ); the CTC region ( $F < 0$ ) is the *supersonic* flow region. The acoustic Hawking temperature

$$T_H^{\text{ac}} = \kappa_{\text{ac}}/(2\pi), \quad \kappa_{\text{ac}} = \frac{1}{2}|dF/dr|_{r=r_H} \quad (8)$$

equals the gravitational Hawking temperature at the LP chronology horizon to machine precision (relative difference  $< 10^{-12}$ ; `compare_acoustic_vs_gravitational_T_H`).

**Experimental access.** The acoustic-analog mapping is the *only currently lab-tractable path* to the Systrophē physics. In a Bose–Einstein condensate (BEC) with a quantum vortex of winding  $\ell$ , the azimuthal flow  $v_\phi(r) = \hbar/(mr)$  automatically violates  $v_\phi(R) > c/2$  once  $R < 2\xi_{\text{BEC}}$  (the healing length), reproducing the Tipler supercritical condition  $\omega R > 1/2$ . A  $\mathbb{Z}_3$  triple-vortex configuration realises the Möbius cover. The full design proposal is in `docs/EXPERIMENTAL_ACOUSTIC_AN`.

## 5 Floquet engineering and discrete-scale invariance

### 5.1 Joint Floquet on $(t, b)$

Module `floquet_mobius` computes Floquet quasi-energies on the joint time-circle  $\times \mathbb{Z}_3$  branch space. The three-level Hamiltonian

$$H_{\text{static}} = \text{diag}(e_0, e_1, e_2) + g \cdot (|0\rangle\langle 1| + |1\rangle\langle 2| + |2\rangle\langle 0| + h.c.) \quad (9)$$

is driven by  $V(t) = A \sin(\Omega t)(S + S^\dagger)$ , where  $S$  is the  $\mathbb{Z}_3$  cycle-shift. Quasi-energies satisfy  $\mathbb{Z}_3$  cyclic-permutation symmetry; resonances open at  $\Omega = e_b - e_{b'}$ .

The `floquet_engineering` extension module maps the (amplitude, frequency) plane and identifies the CTC stability gap

$$\Delta_{\text{stable}}(\text{amp}, \Omega) = \min_{i \neq j} |\epsilon_i - \epsilon_j|/\Omega, \quad (10)$$

quantifying drive-induced stabilisation or destabilisation of CTC bands.

### 5.2 Tipler-sinusoid discrete-scale invariance and cascade-DSI

The Tipler sinusoid  $F(r) = \cos(\alpha \ln r/R + \delta)$  has zero set  $\{r_n = r_0 e^{\pi n/\alpha}\}$ : a geometric progression with ratio

$$\rho = e^{\pi/\alpha}, \quad (11)$$

verified to machine precision in `base_tipler_sinusoid_is_dsi`. This is the textbook DSI signature of a log-periodic precursor (Sornette 1998). A multi-cylinder cascade

$$F_{\text{cascade}}(r) = \sum_{k=0}^{L-1} A_k \cos(\alpha_k \ln r/R + \delta_k), \quad \alpha_k = \alpha_0 \lambda^k, \quad (12)$$

extends to non-trivial box-counting dimension (measured  $> 0.3$  in `cascade_box_dimension`). `dsi_observables` ports this machinery to data analysis: log-periodic precursor fits and Lomb–Scargle log-frequency search.

## 6 Back-reaction and self-consistency

The `back_reaction` module wires `hadamard_offtrace` and `newton_kantorovich` into a composite back-reaction residual

$$\mathcal{R}(\delta) = w_T \sum_r \|\langle T_{\mu\nu} \rangle(s_1; r)\| + w_L \sum_r |L_{\text{pair}}(\delta; r)|. \quad (13)$$

For the matched pair (identical  $a, R, A$ ), the residual is minimised at  $\delta = \pi$ , matching the v0.11 CTC log-measure result. This is *not* a full Einstein-coupled back-reaction solve (the spatial metric is not updated between iterations); it is a static residual minimisation. Full back-reaction requires an ADM-coupled solver, which is enabled by the `adm_export` module providing initial-data hand-off for Einstein Toolkit / similar NR codes.

## 7 D-CTC and quantum information

Module `d_ctc` implements Deutsch’s 1991 CTC fixed-point equation

$$\rho_{\text{CTC}} = \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho_{\text{CTC}})U^\dagger] \quad (14)$$

on the  $\mathbb{Z}_3$  branch space as the natural 3-dimensional CTC register. The cyclic-shift unitary  $U = R_{\text{CR}} \otimes S$  admits the maximally-mixed CTC state as a Banach fixed point; the iteration converges to machine precision in tens of iterations.

This puts SYSTROPHE in conversation with the Aaronson–Watrous (2008) PSPACE result on D-CTC complexity: the underlying spacetime geometry is here concretely instantiated.

## 8 Validation modules for previously speculative items

Six items previously catalogued as “open speculative” in `docs/INTERPRETATIONS.md` have been promoted to concrete calculations in v0.15.0. Each item now has a validation module that fixes the missing input and produces a quantitative answer.

**I.1:  $\mathbb{Z}_3$  cover as wormhole throat (`wormhole_throat.py`).** The LP angular metric admits an effective Morris-Thorne shape function  $b_{\text{eff}}(r) = r - L(r)/r$ , with throat candidates at  $L(r_t) = 0$  (the CTC band boundary) and flaring-out  $b'(r_t) < 1$  at the outer boundary of each CTC band. However, a genuine Morris-Thorne wormhole connects two asymptotic regions through a topological *junction*; the LP exterior is a *single connected* spacetime with  $L = 0$  a coordinate locus, not a junction. Direct evaluation of  $R_{\mu\nu}$  at  $L = 0$  returns vacuum. **Verdict:** the kinematic shape-function mapping is well-defined, but the  $L=0$  locus is *not* a Morris-Thorne throat — it is a vacuum-supported CTC band boundary in a single-connected manifold.

**I.2: Casimir vs. exotic matter (`exotic_matter_accounting.py`).** The LP  $L = 0$  locus is Einstein vacuum (item I.1) and requires *no* local exotic matter; the geometry is sourced remotely by the rotating cylinder interior (positive-energy van Stockum dust). The “Casimir-replaces-exotic-matter” question is therefore vacuous for this geometry. Under the hypothetical that one nevertheless interpreted  $L = 0$  as a Morris-Thorne junction: equating the MT exotic budget  $|\rho_{\text{exotic}}|(r_t) = b'(r_t)/(8\pi r_t^2)$  to the Casimir budget  $|\rho_C| = \pi^2/(720d^4)$  gives  $d_{\text{req}}/r_t \sim 0.7$  at  $r_t \sim 1$  (natural units), so the cavity does not geometrically fit. **Verdict:** the LP throat stays open without exotic matter because no Morris-Thorne wormhole is being built; the conditional Casimir/exotic-matter comparison gives quantitative NO regardless.

**I.3: Self-consistent  $\delta$  (`chronology_protection.py`).** Newton-Kantorovich on the composite back-reaction residual  $F(\delta) = w_T \sum_r |\langle T_{\mu\nu} \rangle| + w_L \sum_r |L_{\text{pair}}|$  from generic initial deltas converges to a  $\delta$  in the half-circle containing  $\pi$  for the matched-pair case. **Verdict: consistent** with chronology-protection conjecture.

**I.4: Möbius monodromy as U(1) Wilson loop (`wilson_loop.py`).** The flat connection  $A_\phi = (\gamma_{\text{eff}} + 2\pi b/3)/(2\pi)$  has Wilson loop  $W = \exp(i(\gamma_{\text{eff}} + 2\pi b/3))$ , reproducing the  $\mathbb{Z}_3$  branch phase. Field strength  $F = dA = 0$  everywhere; Chern number  $c_1 = 0$ . The non-trivial content is the holonomy classifying  $\pi_1(S^1) = \mathbb{Z}$ . **Verdict: concrete connection constructed.**

**I.5: DCE photon flux (`dynamical_casimir.py`).** Cavity with modulation  $d(t) = d_0(1 + \epsilon \sin \Omega t)$  produces  $\langle N \rangle_{\text{resonant}} = \sinh^2(\epsilon \omega_n t/4)$  on resonance and  $\langle N \rangle_{\text{off}} = \epsilon^2/(Q^2 \delta^2)$  off resonance, where  $\delta = |\Omega/\Omega_n - 1|$ . The regime selector picks the right formula by detuning vs. linewidth  $1/Q$ . **Verdict: testable for any  $(Q, \epsilon, \delta)$ .**

**I.6: Natural QFT vacuum (`vacuum_states.py`).** Three candidates implemented: Boulware (well-defined for  $F > 0$ ), adiabatic (well-defined for  $|dF/dr|/|F| < 1$ ), and Hartle-Hawking-analog (requires Killing horizon  $\alpha^2 = F + K^2/L = 0$ ). For supercritical LP, the chronology horizon ( $F = 0$ ) is generically *not* a Killing horizon since  $K^2/L \neq 0$  at  $F = 0$ . **Verdict: NO.** The natural HH-analog vacuum does *not* exist on the generic supercritical LP exterior; the cavity-Casimir state cannot be the natural vacuum in the standard QFTCS sense. Use Boulware or adiabatic instead.

The most consequential resolution is the I.1+I.2 pair: the LP “throat” ( $L = 0$  locus) is vacuum-supported, single-connected, and not a Morris-Thorne junction. The throat stays open *without* exotic matter, because no Morris-Thorne wormhole exists in this geometry. Popular informal framings that treated the  $L = 0$  surface as a Morris-Thorne junction (and asked whether Casimir could supply the resulting exotic-matter budget) are vacuous for the actual Systrophe geometry. I.6 is also negative: the Hartle-Hawking-analog vacuum does not exist generically on the supercritical LP background. The other three items (I.3, I.4, I.5) are constructive.

## 9 Phase 2a: quantitative chronology protection (v0.20.0)

The 4D vacuum trace anomaly  $K_{\text{Kretsch}}/(2880\pi^2)$  computed in Section ?? is a *local* geometric quantity that remains bounded at the Cauchy horizons of the supercritical Tipler exterior (the Kretschmann scalar is finite at the simple zeros of  $F$ ). One could mistakenly conclude that the chronology-protection conjecture is therefore inactive on this spacetime. v0.20.0 closes that gap.

### 9.1 2D Polyakov stress tensor on the radial-temporal sector

We treat the  $(t, r)$  sector of the Lewis–Papapetrou exterior as a 2D conformally-flat spacetime with conformal factor  $e^{2\sigma} = F(r)$  ( $\sigma = \frac{1}{2} \ln F$ ). Following the Davies–Fulling–Unruh / Christensen–Fulling 2D Polyakov formalism [9, 10, 11], the renormalised stress tensor of a massless conformally-coupled scalar in the Boulware vacuum is, in tortoise coordinates  $dr_* =$

$\sqrt{h/F} dr,$

$$\langle T_{tt} \rangle_B = \frac{1}{24\pi} (\partial_{r_*} \sigma)^2, \quad (15)$$

$$\langle T_{r_* r_*} \rangle_B = \frac{1}{24\pi} [(\partial_{r_*} \sigma)^2 - 2\partial_{r_*}^2 \sigma], \quad (16)$$

$$\langle T_{tr_*} \rangle_B = 0, \quad (17)$$

$$\langle T^\mu{}_\mu \rangle_B = R_{2D}/(24\pi) \quad (\text{Polyakov trace identity}). \quad (18)$$

The Hartle–Hawking and Unruh analogs add constant thermal offsets  $\pi T_H^2/12$  to the lightcone components  $T_{uu}, T_{vv}$  (symmetric for HH, one-sided for Unruh).

## 9.2 Measured chronology-protection divergence

Near a simple zero  $r_H$  of  $F$  with  $F(r) \approx F'_H(r - r_H)$ , the Boulware components diverge as

$$\langle T_{tt} \rangle_B \sim \frac{F'_H}{96\pi(r - r_H)}, \quad \langle T_{rr} \rangle_B \sim -\frac{F''_H}{24\pi F'_H(r - r_H)^2}. \quad (19)$$

A log–log fit on a 20-point geometrically-spaced sweep at the first three Cauchy horizons of  $\omega = 2$ ,  $R = 1$  (so  $\alpha = \sqrt{15}$ ,  $r_H \in \{1.405, 3.163, 7.118\}$ ) gives:

| horizon          | $\langle T_{tt} \rangle$ power | $\langle T_{rr} \rangle$ power | fit rms              |
|------------------|--------------------------------|--------------------------------|----------------------|
| $r_{H1} = 1.405$ | −1.007                         | −1.998                         | $5.3 \times 10^{-3}$ |
| $r_{H2} = 3.163$ | −0.997                         | —                              | $1.6 \times 10^{-3}$ |
| $r_{H3} = 7.118$ | −1.001                         | —                              | $8.6 \times 10^{-4}$ |

All three match the analytic prediction to better than 1% in power, and the Polyakov trace identity holds at  $2.2 \times 10^{-16}$  (machine precision) across the 20-point sweep.

A sweep across nine values of the rotation parameter  $a = \omega R$  from 0.6 to 3.0 gives mean power  $-1.0047 \pm 0.0012$  for  $\langle T_{tt} \rangle$  and  $-1.9987 \pm 0.0005$  for  $\langle T_{rr} \rangle$ . The divergence rate is *geometric*: it depends only on the existence of a simple  $F$ -zero, not on the log-frequency  $\alpha = \sqrt{4a^2 - 1}$ .

## 9.3 Pair extension: anti-phase extinction also kills the divergence

For the matched **SystrophePair** with relative phase  $\delta \in [0, \pi]$ , the linearised joint  $F_{\text{pair}}(r; \delta) = 2 \cos(\delta/2) \cdot F_1(r; \delta/2)$  shares the same simple-pole structure for any non-zero amplitude. At  $\delta = \pi$ ,  $F_{\text{pair}} \equiv 0$  identically: zero Cauchy horizons, zero Polyakov  $\sigma$ , zero divergence. The *operational* chronology off-switch (anti-phase pair, v0.1+) therefore extinguishes the QFTCS back-reaction signal at the same time as it extinguishes the classical CTCs — both manifest the same  $F_{\text{pair}} = 0$  degeneracy.

## 9.4 Reconciliation with the 4D Kretschmann finiteness

SYSTROPHE v0.10.0 had verified that the 4D Kretschmann scalar (and hence the local trace anomaly  $K/2880\pi^2$ ) remains bounded at the first Cauchy horizon. The Phase 2a result does not contradict that. The 4D trace anomaly is the *state-independent* geometric piece of  $\langle T^\mu{}_\mu \rangle$ ; the *state-dependent* Polyakov contribution on the  $(t, r)$  sector — which is the back-reaction that drives Einstein’s equations — is what diverges. Hawking’s conjecture concerns the latter, not the former.

**Verdict.** SYSTROPHE v0.20.0 verdict: `chronology_protection_consistent`. The full multi-horizon, multi-state report is reproduced from a single command: `python examples/phase_2a_chronology_pr` in  $\sim 0.1$  s.

## 10 Summary and outlook

SYSTROPHE v0.20.0 provides the smallest open-source toolkit known to the author that (i) constructs an exact closed-form CTC spacetime, (ii) computes renormalised quantum-field-theoretic observables on it, (iii) verifies Hawking’s chronology-protection conjecture quantitatively against three independent Cauchy horizons, and (iv) supports both an experimental analog (BEC vortices) and a quantum-information substrate (D-CTC).

**Tests.** The package now contains 1321 passing tests covering all numerical claims in this paper, including the 18 new Phase 2a tests in `tests/test_stress_energy_ctc.py`. The full source is at [github.com/Zynerji/systrophe](https://github.com/Zynerji/systrophe); license is MIT.

**Next steps.** (a) BEC-vortex experimental design collaboration (Tier 1); (b) full back-reaction NR run via ADM hand-off; (c) extension to the multi-cylinder cascade DSI as a model of analog-cosmological log-periodic precursors; (d) Phase 2b: a 4D mode-sum Hadamard subtraction giving the state-dependent off-trace  $\langle T_{\mu\nu} \rangle$ .

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